PG No. 36 $D_{3\mathrm{d}}(1)$ $\bar{3}m$ (-31m setting) [trigonal]

Table 1: Representation matrices

Irrep.						
A_{1g}	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}1\end{bmatrix}$	$2_{120}:\begin{bmatrix}1\end{bmatrix}$	$2_{210}:\begin{bmatrix}1\end{bmatrix}$
	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}1\end{bmatrix}$
A_{2g}	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}-1\end{bmatrix}$	$2_{120}:\begin{bmatrix}-1\end{bmatrix}$	$2_{210}:\begin{bmatrix}-1\end{bmatrix}$
	$-1:\begin{bmatrix}1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}-1\end{bmatrix}$
E_g	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$2_{120}:\begin{bmatrix}-1&0\\0&1\end{bmatrix}$	$2_{210}:\begin{bmatrix}\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}-1&0\\0&1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$
A_{1u}	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}1\end{bmatrix}$	$2_{120}:\begin{bmatrix}1\end{bmatrix}$	$2_{210}:\begin{bmatrix}1\end{bmatrix}$
	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}-1\end{bmatrix}$
A_{2u}	$1:\begin{bmatrix}1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}1\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}1\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}-1\end{bmatrix}$	$2_{120}:\begin{bmatrix}-1\end{bmatrix}$	$2_{210}:\begin{bmatrix}-1\end{bmatrix}$
	$-1:\begin{bmatrix}-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}-1\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}-1\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}1\end{bmatrix}$
E_u	$1:\begin{bmatrix}1&0\\0&1\end{bmatrix}$	$3^+_{001}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$3^-_{001}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$2_{1-10}:\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$	$2_{120}:\begin{bmatrix}-1&0\\0&1\end{bmatrix}$	$2_{210}:\begin{bmatrix}\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&-\frac{1}{2}\end{bmatrix}$
	$-1:\begin{bmatrix}-1&0\\0&-1\end{bmatrix}$	$-3^+_{001}:\begin{bmatrix}\frac{1}{2}&\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$-3^-_{001}:\begin{bmatrix}\frac{1}{2}&-\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{1-10}:\begin{bmatrix}-\frac{1}{2}&\frac{\sqrt{3}}{2}\\\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$	$\mathrm{m}_{120}:\begin{bmatrix}1&0\\0&-1\end{bmatrix}$	$\mathrm{m}_{210}:\begin{bmatrix}-\frac{1}{2}&-\frac{\sqrt{3}}{2}\\-\frac{\sqrt{3}}{2}&\frac{1}{2}\end{bmatrix}$